

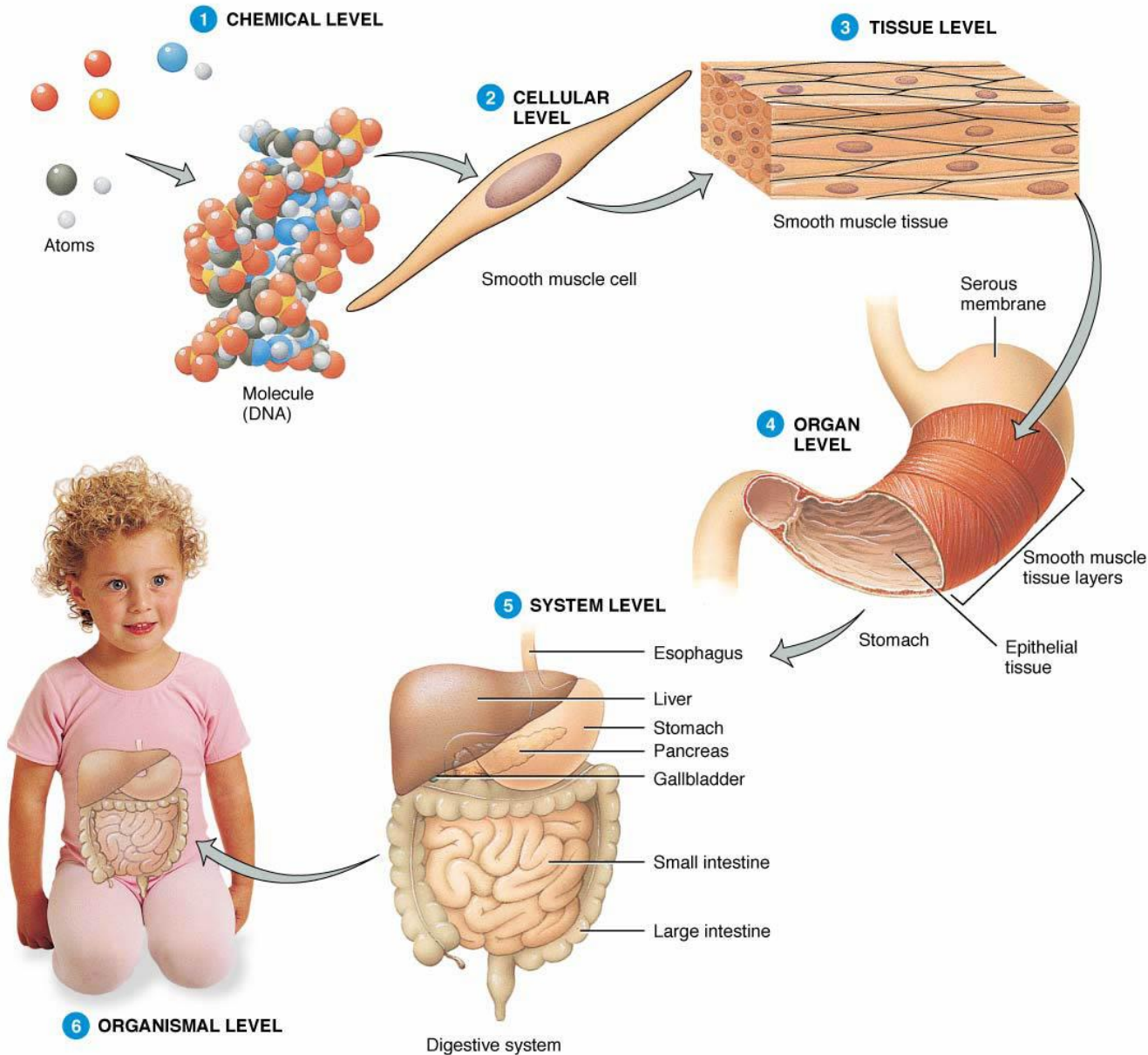


Introduction:

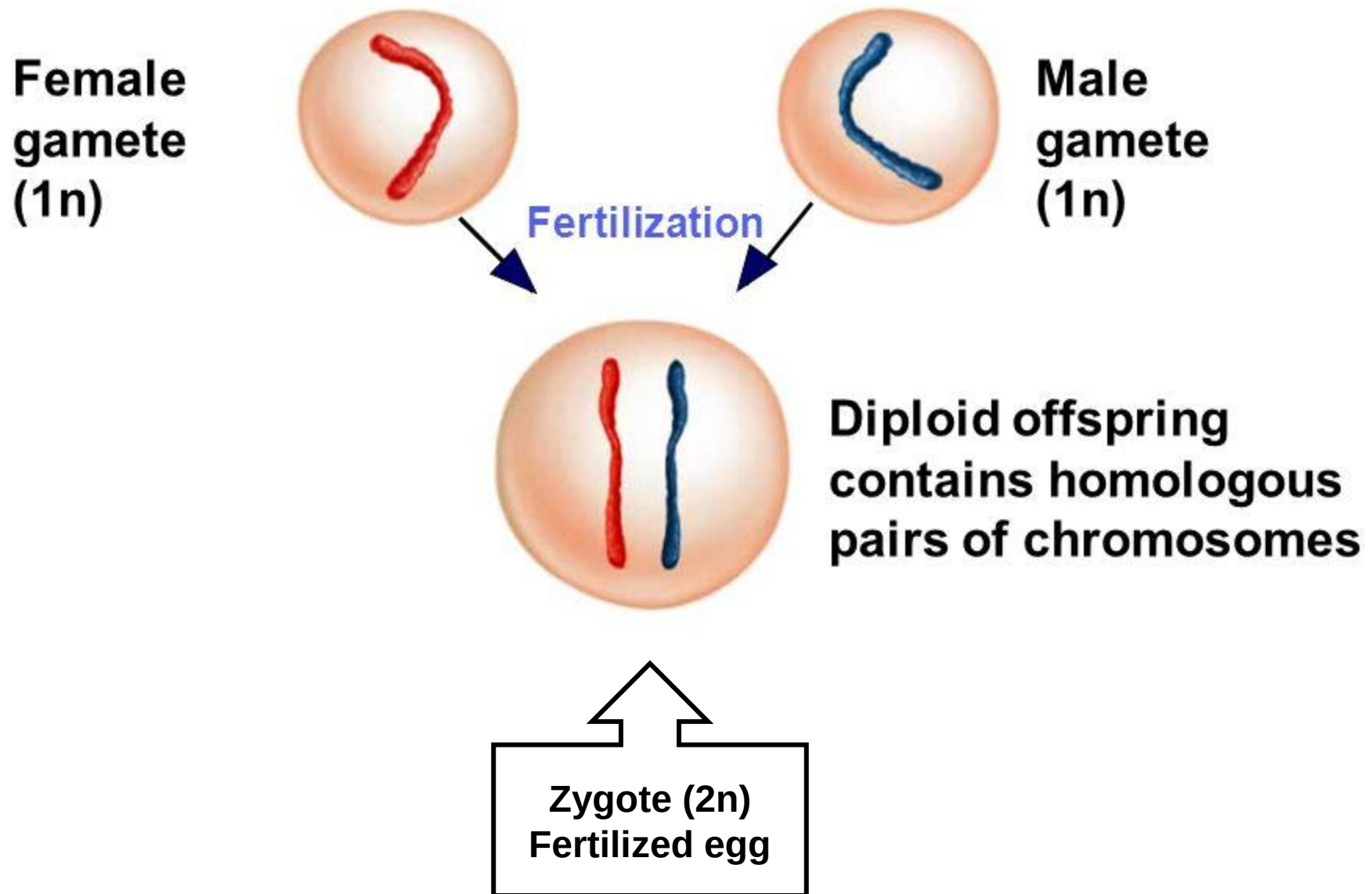
Blood and its components

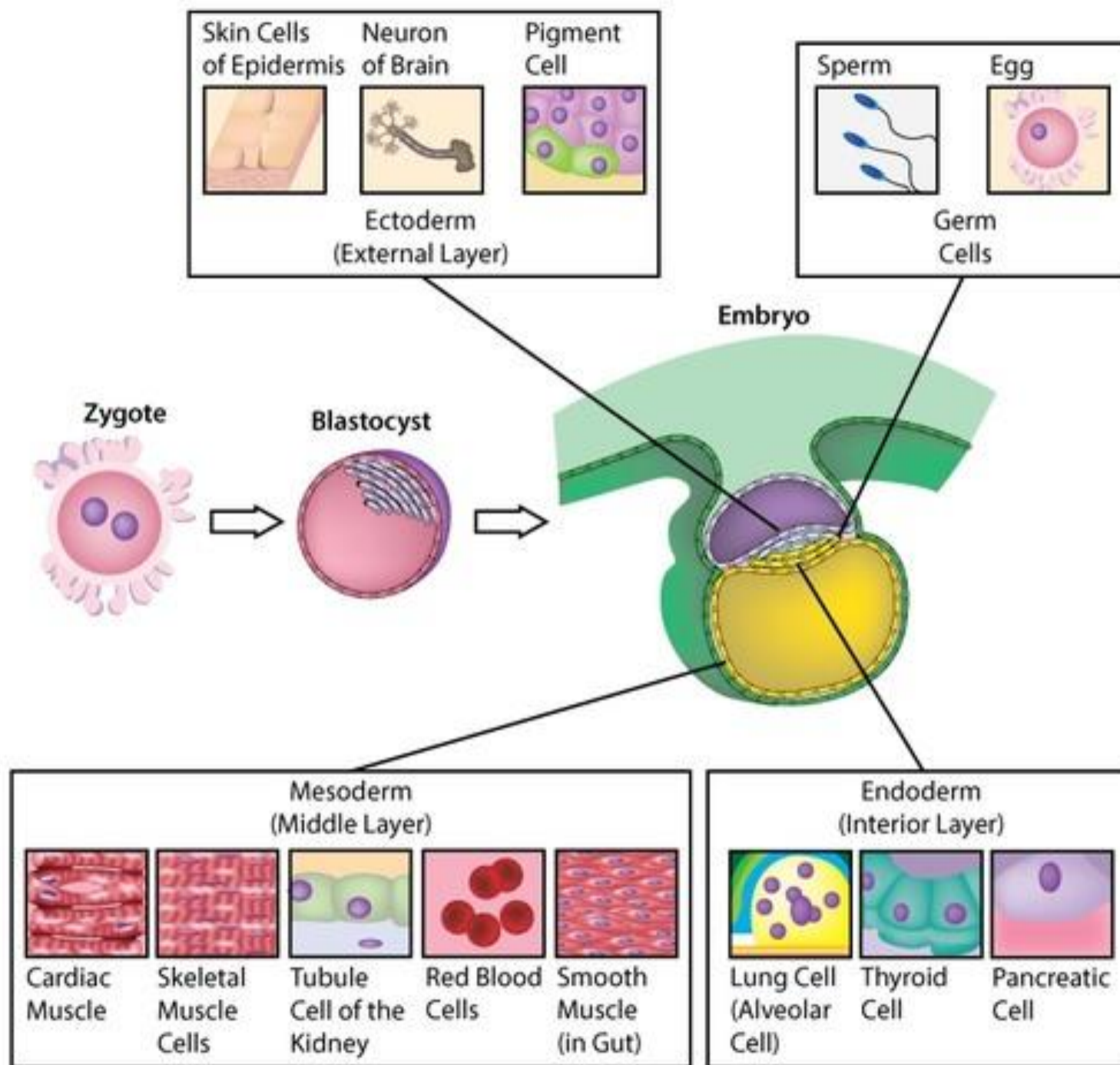
Dr. Jehad Alhmoud

Levels of Organization in the body



- Chemical
- Cellular
- Tissue
- Organs
- System level
- Organismic level





Hematology (Haematology)

- Hemato: blood, ology: study of.
- Is the study of blood (definition, composition, components, functions, and disorders).
- Blood: is the red fluid pumped from the heart and circulated in the blood vessels (arteries, veins, and capillaries).

- It comprises blood cells (formed elements) suspended in plasma.
- Cell: is the basic structural and functional unit of the human body.
- Cell has 3 components: plasma membrane, nucleus, and cytoplasm.

Blood Characteristics

- The average volume of blood in a normal adult is 5 ± 1 liters.
- The total blood volume in a normal adult is 70 mL/kg.
- Blood accounts for 7-8% of the human body weight.
- The color of the blood is bright red in the arteries and dull red in the veins.

- Blood pH is regulated to stay within the narrow range of 7.35 to 7.45, making it slightly basic (alkaline). Blood that has a pH below 7.35 is acidic, whereas blood pH above 7.45 is basic.
- Blood has an average density of approximately $1,060 \text{ kg/m}^3$ (1.060 g/cm^3), which is very close to the density of pure water ($1,000 \text{ kg/m}^3$).

Blood Components

The components of blood are :

➤ **Blood cells (cellular or formed elements), which include:**

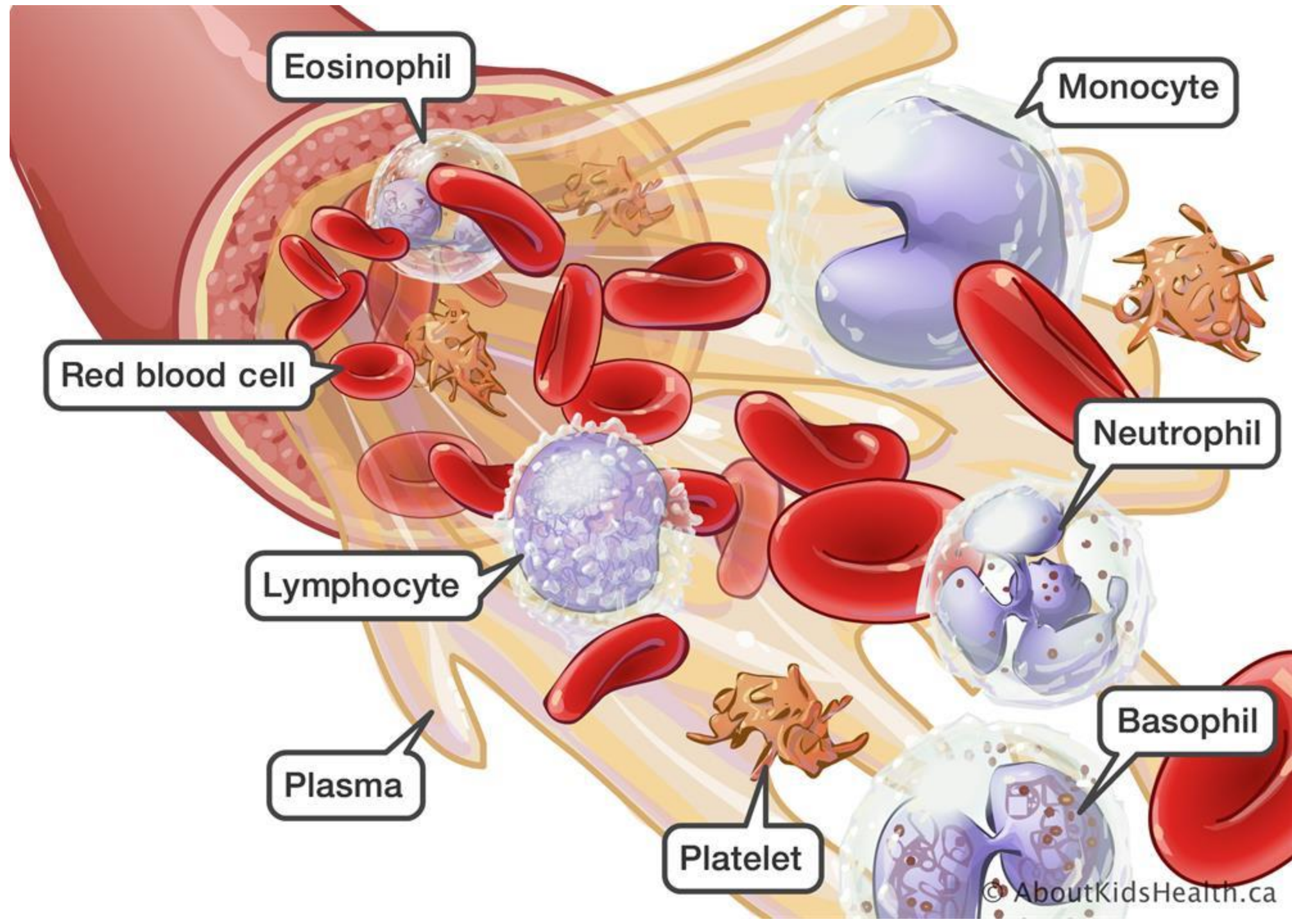
- 1- Red blood cells (RBCs or erythrocytes),
- 2- White blood cells (WBCs or leukocytes) and
- 3- Platelets (plts or thrombocytes).

➤ **Plasma.**

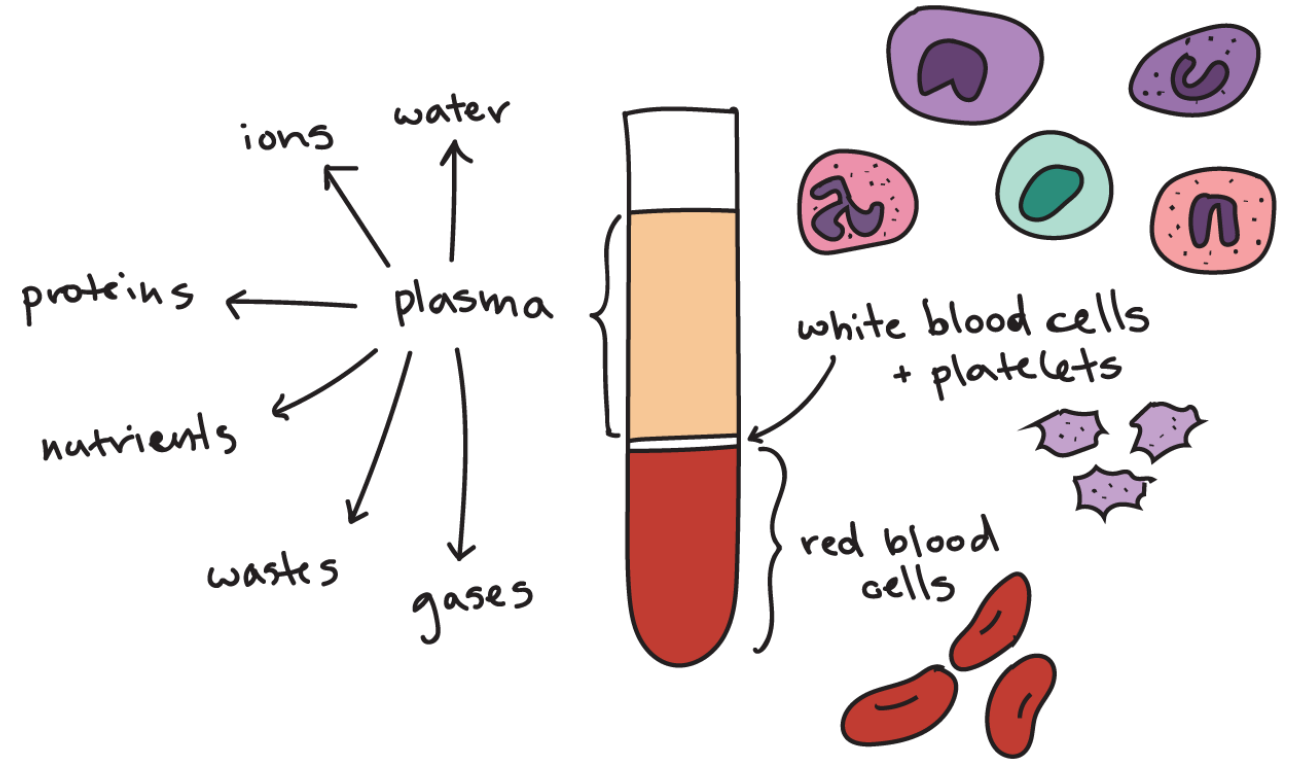
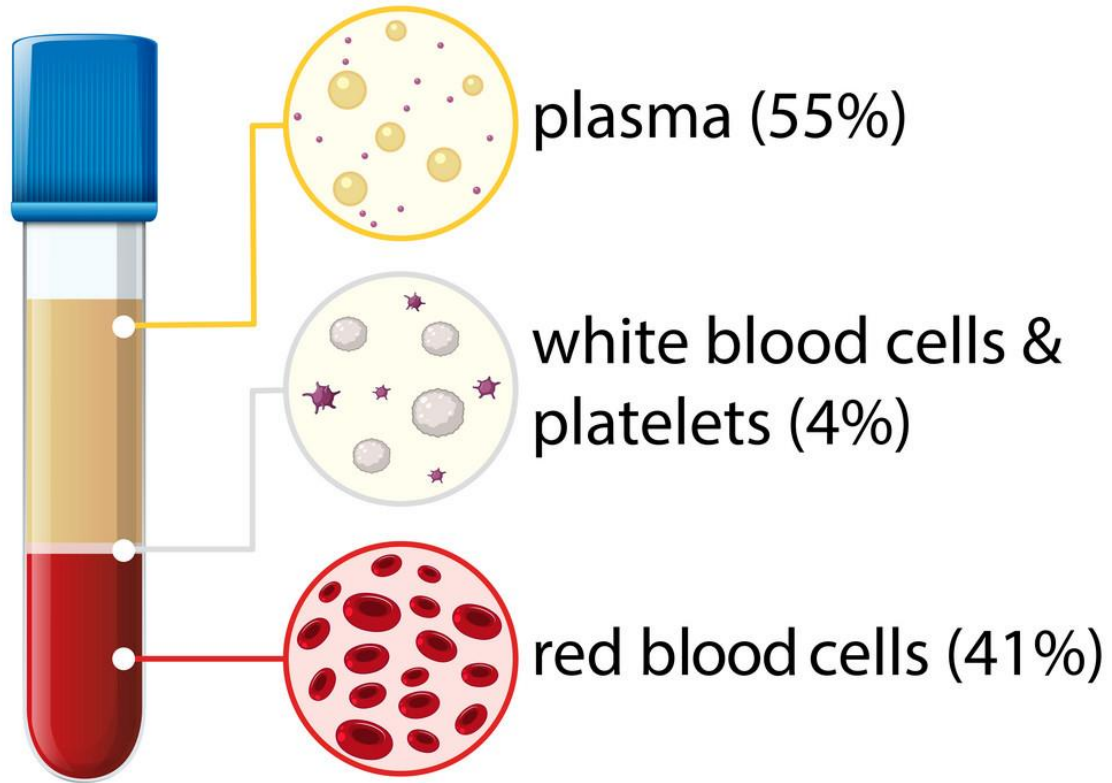
- The plasma portion composes about 55% of the total blood volume, and the cellular elements compose about 45% of the total blood volume.
- The RBCs (erythrocytes) account for about 44%, while the WBCs (leukocytes) and plts (thrombocytes) account for 1% of the total blood volume.

Plasma

- Is the liquid part of the blood in which the blood cells are suspended and many substances (e.g. minerals) are transported.
- It is composed mainly of water (90%) in which proteins, glucose, minerals, hormones, and other substances are dissolved or bound to plasma proteins.
- Contains all coagulation factors, while serum represents plasma from which coagulation factors especially fibrinogen is consumed in blood clotting.
- **Serum = plasma – fibrinogen.**
- **Plasma = serum + fibrinogen.**



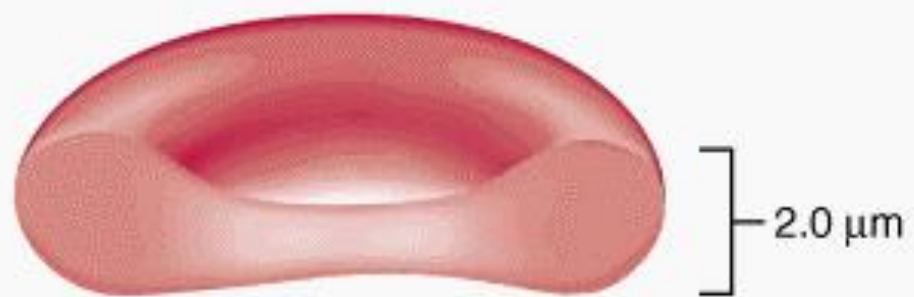
Composition of Blood



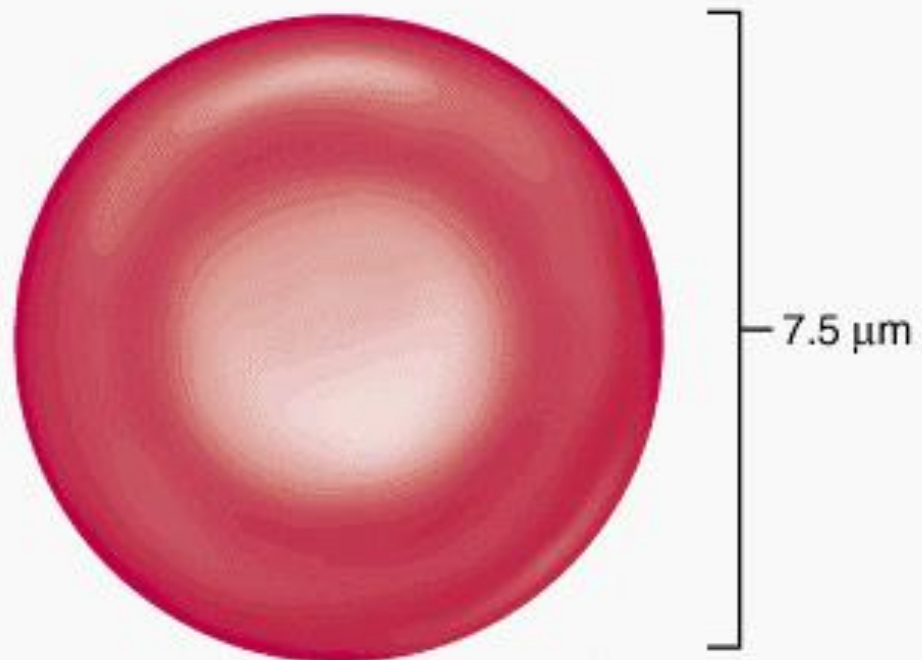
Cellular elements

1- RBCs or Erythrocytes:

- Are the most abundant or numerous cells in the blood.
- They contain hemoglobin, an iron-containing protein, which facilitates O₂ transport.
- Play an important role in gas exchange (O₂ & CO₂).
- Count: 4.5 to 6.0 million (male), 4.2 to 5.4 million (female) per cubic mm (mm³) or microliter (μL).
- Have no nucleus (Anucleate).
- Biconcave disc.
- 7-8 μm in diameter.
- Have a life span of 120 days.



Side view



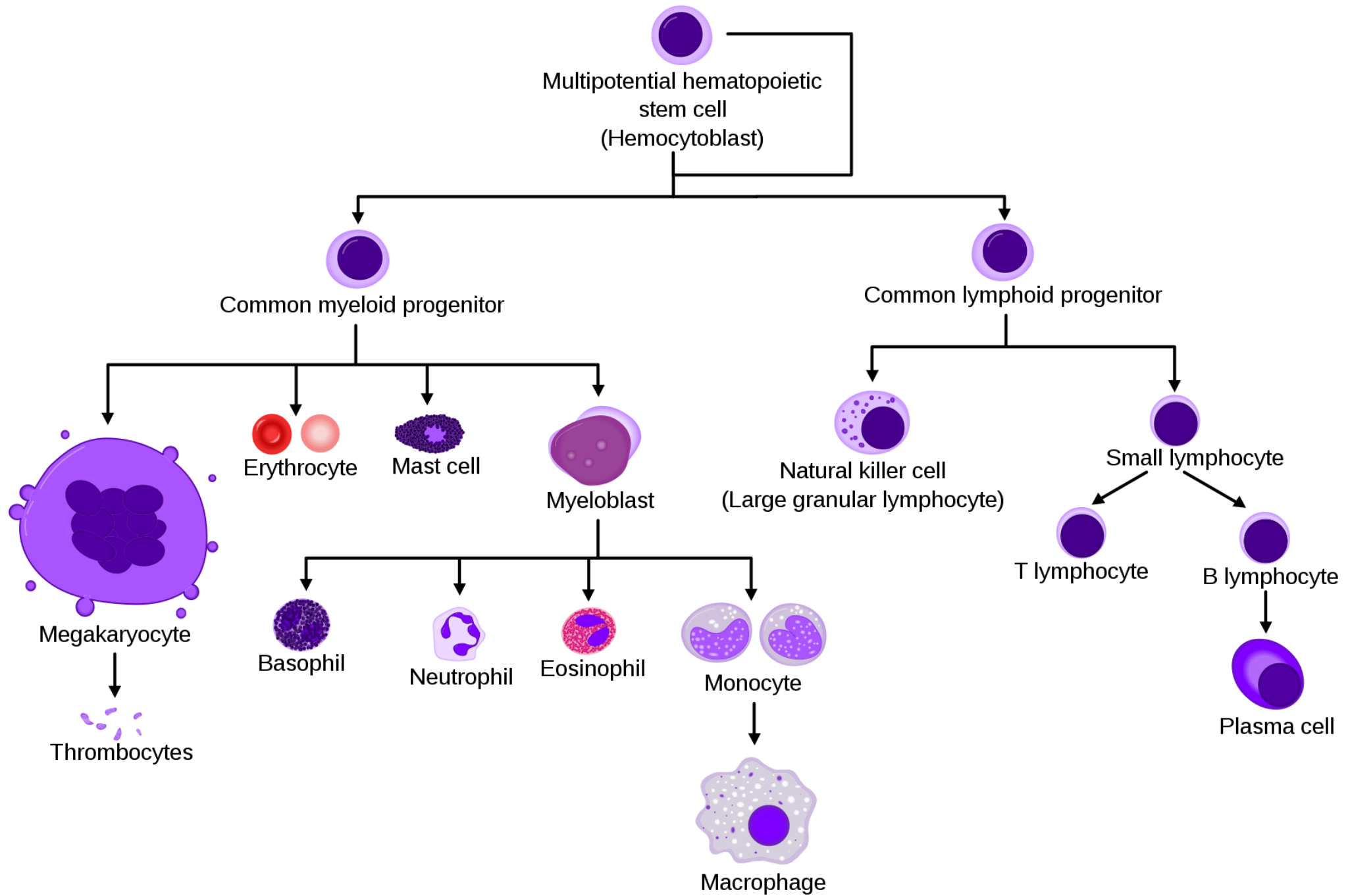
Top view

2- Platelets or thrombocytes:

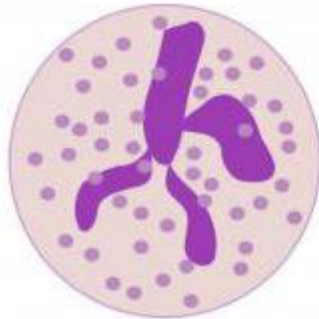
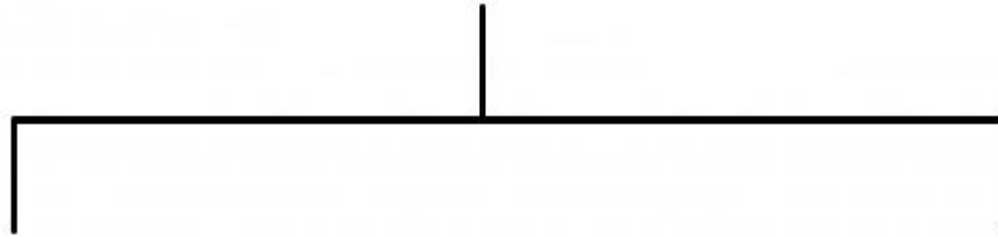
- The 2nd most numerous blood cells.
- Function in hemostasis (prevent bleeding and help stop it when occurs).
- Count: $(150 - 400) \times 10^3$ per cubic mm (mm^3) or microliter (μL).
- They are cellular fragments.
- Are anucleate discs with a diameter of $2.5 \mu\text{m}$.

3- WBCs or leukocytes:

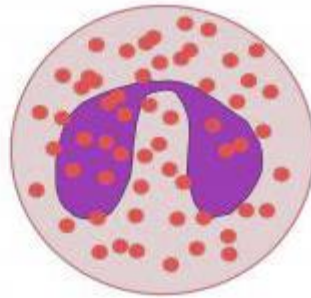
- Are the least numerous and the largest blood cells.
- Count: 4,000 – 11,000 per cubic mm (mm³) or microliter (μL).
- Function in body immunity and defense against foreign substances.
- There are many types of WBCs.
- Leukocytes are usually divided into **granulocytes**, which have specific granules, and **agranulocytes**, which lack specific granules.
- **Granulocytes** involve three main types of cells: neutrophils, eosinophils, and basophils.
- **Agranulocytes** involve two main types of cells: lymphocytes and monocytes (macrophages into tissues).



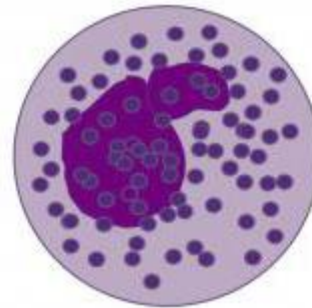
Granulocytes



Neutrophil



Eosinophil

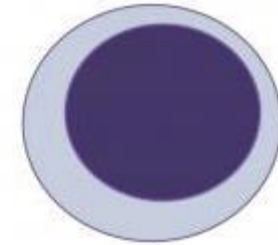


Basophil

Agranulocytes



Monocyte



Lymphocyte

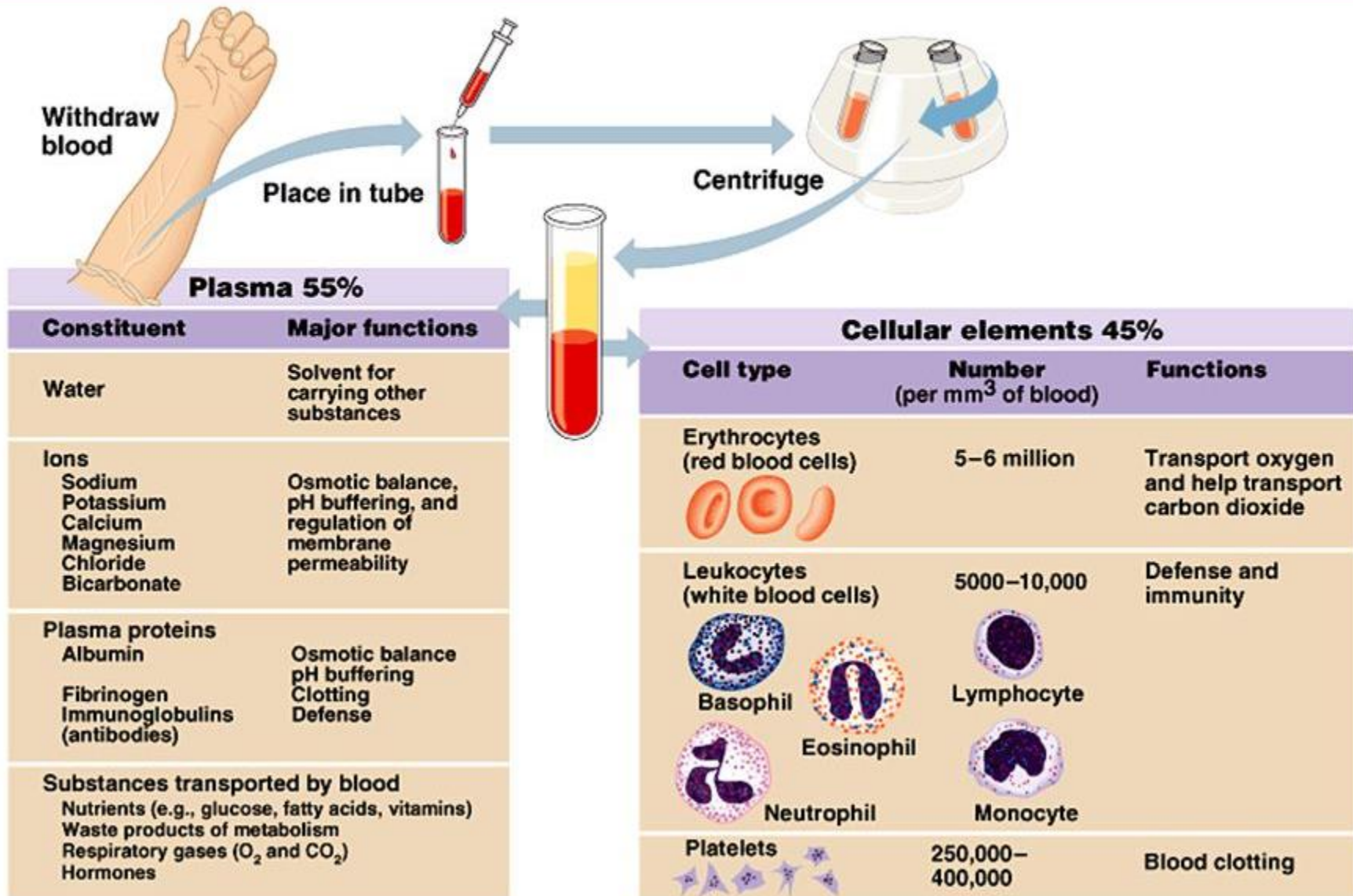
Blood Functions

➤ **Blood can perform many functions:**

- 1) Gas exchange:** RBCs carry the O_2 from the lungs to the cells and tissues and CO_2 from the cells and tissues to the lungs to be exhaled.
- 2) Defense** against foreign substances and materials such as bacteria, viruses, toxins and others. This function is performed by WBCs.
- 3) Blood clotting and coagulation** by platelets and coagulation factors to prevent bleeding or hemorrhage from occurrence or stop it when occurs.

- 4) **Transport** of nutrients such as proteins, glucose, fatty acids, and others to the cells (Transport function).
- 5) **Removal** of waste substances such as CO₂, urea, and bilirubin.
- 6) **Regulation** of body temperature.
- 7) **Buffering** (pH maintenance).
- 8) **Regulation** of osmotic pressure.

BLOOD COMPONENTS



Summary

- The cell is the basic structural and functional unit of the human body.
- Hematology is the science that deals with the study of blood.
- Blood, the fluid that circulates in the blood vessels, is composed of two components: the formed or cellular elements and plasma.
- The formed or cellular components of blood represent 45% of the blood and include; red blood cells, white blood cells, and platelets.

- White blood cells are either granulocytes (neutrophils, eosinophils, and basophils) or agranulocytes (lymphocytes and monocytes).
- Plasma, which represents 55% of the blood, is the fluid part of the blood in which the formed elements are suspended and the nutrients and substances are transported.
- Blood performs many functions including gas exchange, defense and immunity, blood hemostasis, transport of nutrients and substances, and regulation rule.